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CRITICAL ANALYSIS REPORT ON THE 2021 PROPERTY PRICE REGISTER DATASET

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1 Introduction & Dataset

This report presents a critical evaluation of a dashboard developed using the 2021 Residential Property Price Register (PPR) dataset. The Residential Property Price Register is officially published by the Property Services Regulatory Authority (PSRA) under Section 86 of the Property Services (Regulation) Act 2011. The dataset records all residential property transactions in Ireland since 1 January 2010, based on declarations submitted to the Revenue Commissioners for stamp duty purposes. It provides detailed transactional information including date of sale, sale price, property address, county, and classification of dwelling type (new or second-hand).

The PPR dataset is particularly valuable for housing market analysis because it is comprehensive, official, and transaction-based rather than survey-based. Transaction-level data reduces estimation bias and enhances reliability, making it suitable for descriptive and exploratory analytics. In addition, the dataset's longitudinal structure allows temporal analysis, while the inclusion of geographic identifiers (counties) enables spatial comparisons.

Building on these characteristics, the dashboard was developed under the Housing theme with the primary objective of examining regional price disparities and patterns in sales activity across Ireland. By focusing on key variables such as date of sale, price, county, and dwelling type; the analysis shows spatial inequality in property values and identifies fluctuations in transaction volumes over time. This dual focus on geography and temporality aligns with established housing market research approaches, where both location and timing are central determinants of price behaviour (Goodman & Thibodeau, 2008; Soltani et al., 2021).

The dashboard therefore aims not merely to present data, but to transform raw transactional records into interpretable insights for policymakers, investors, and the general public. Through carefully selected visualizations, it enables pattern detection,

ranking, and comparison, which are central goals of effective data communication (Few, 2012; Cairo, 2016).

2 Design Justification and Evaluation

2.1 Spatial Distribution of Property Prices

A choropleth map was selected as the primary geographic visualization because it effectively communicates spatial variation across predefined geographic boundaries. Since the dataset includes a categorical geographic variable (County) and a quantitative variable (average property price), a choropleth map is well-suited for representing aggregated values across regions. Unlike heatmaps, which represent intensity across continuous surfaces and may obscure administrative boundaries, choropleths preserve clear regional demarcations, making them more appropriate for policy-relevant analysis (Brewer, 2015). Similarly, isopleth (contour) maps are designed to show gradual transitions across space and are typically used for phenomena such as temperature or elevation. Housing prices, however, are reported and analyzed within discrete administrative units, making the choropleth a more conceptually accurate choice.

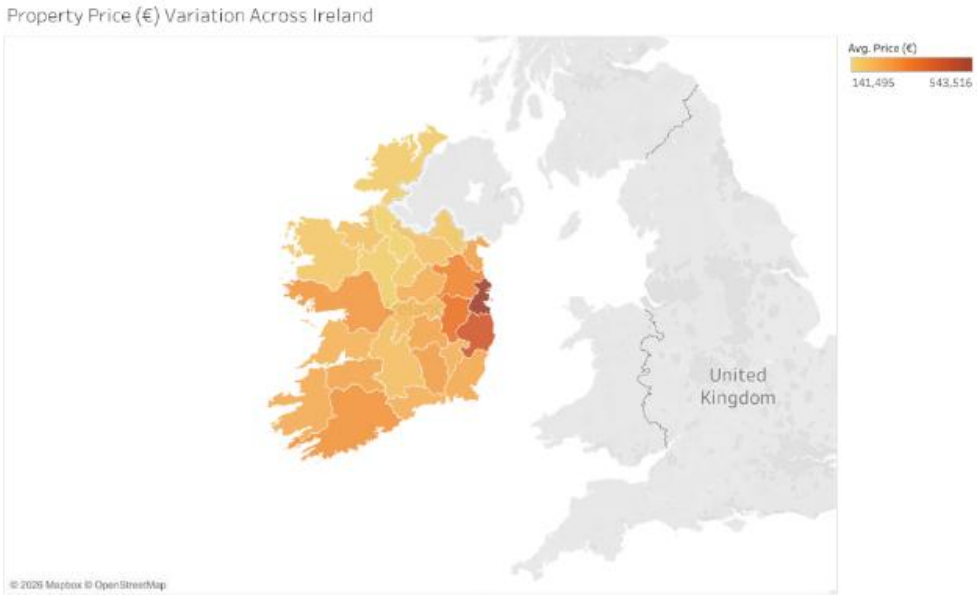


Figure 1 Choropleth map showing variation in property price across Ireland

Cartograms, while useful for emphasizing magnitude by resizing regions according to data values, distort geographic shape and spatial relationships. In the context of housing policy and regional comparison, preserving geographic accuracy is important

for recognizability and interpretability (Slocum et al., 2009). Choropleth maps also leverage preattentive visual processing through colour intensity. Variations in hue or saturation allow viewers to quickly detect high and low values without extensive cognitive effort (Ware, 2013). This supports rapid identification of expensive versus affordable counties, fulfilling the dashboard's objective of highlighting regional disparities.

2.2 Temporal Sales Trend

To visualize changes in transaction volume across the 12-month period, a line chart was employed. Line charts are widely regarded as the most effective tool for representing time-series data because they emphasize continuity and trend progression (Few, 2012). In time-series analysis, the x-axis typically represents chronological progression, while the y-axis represents the measured variable; in this case, number of property transactions. The connected data points in a line chart visually encode direction, slope, and rate of change, enabling viewers to quickly identify upward or downward trends, seasonal patterns, or unusual fluctuations.

An alternative such as a bar chart could display monthly sales volumes, but bars emphasize discrete comparisons rather than continuity. Research in graphical perception shows that viewers interpret slope and direction more intuitively in line charts when analyzing sequential data (Cleveland & McGill, 1984). Because housing transactions occur continuously over time, a line chart more accurately reflects the temporal nature of the dataset.

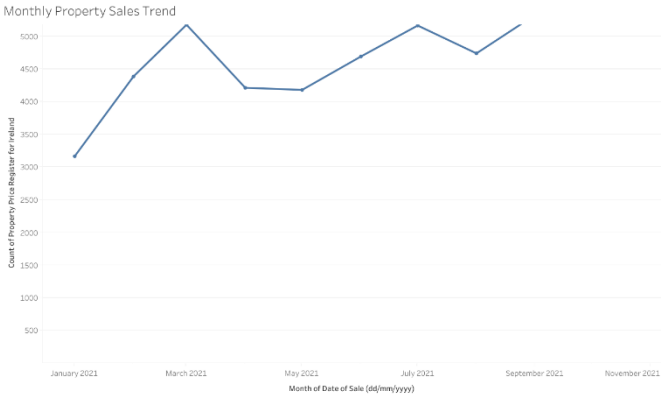


Figure 2 Line chart showing the monthly property sales trend

2.3 New Dwelling vs Second-Hand Dwelling Price

A side-by-side bar chart was used to compare average prices of new dwellings and second-hand dwellings. Bar charts are particularly effective for comparing discrete categories because they rely on length along a common baseline, which is one of the most accurately perceived visual encodings (Cleveland & McGill, 1984). Human perception is significantly more precise when comparing aligned lengths than when comparing angles or areas. For this reason, bar charts are

generally preferred over pie charts for quantitative comparison tasks (Few, 2012). A pie chart would require viewers to estimate angles or relative areas, which introduces perceptual distortion and reduces precision. By placing the two categories side-by-side, the viewer can immediately assess the magnitude of difference between new and second-hand properties. The shared baseline ensures accurate comparison, while the categorical separation preserves clarity. This design decision prioritizes accuracy, simplicity, and interpretability. In addition, bar charts are highly familiar to general audiences, which reduces cognitive load and increases accessibility; an important consideration in public-facing dashboards (Cairo, 2016).

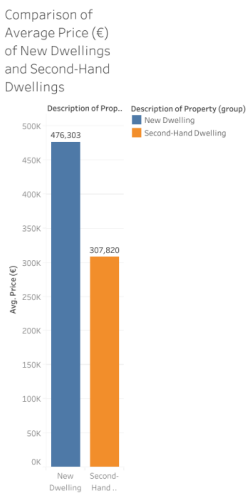


Figure 3 Bar chart showing comparison of average price of second-hand and new dwellings

2.4 Ranking Most Expensive Counties

A horizontal bar chart was used to rank the top ten counties by average property price. Ranking tasks require clear magnitude comparison, and bar charts are widely recognised as optimal for such tasks because length comparison is perceptually reliable (Cleveland & McGill, 1984). The decision to limit the display to the top ten counties reduces visual clutter and focuses attention on the most relevant observations. Information overload can impair comprehension, and selective filtering enhances usability in dashboards (Few, 2012).

A horizontal orientation was chosen because county names can be relatively long. Horizontal bars accommodate longer labels more effectively than vertical bars, avoiding rotated text and improving readability. Research suggests that readability significantly influences user comprehension and engagement in data displays (Ware, 2013). To further enhance interpretability, a colour gradient was applied alongside bar length. This introduces redundant encoding; representing magnitude using both length and colour intensity to which strengthens visual hierarchy and improves pattern recognition

(Cairo, 2016). By reinforcing ranking visually, the chart supports quick identification of market leaders without sacrificing quantitative clarity.

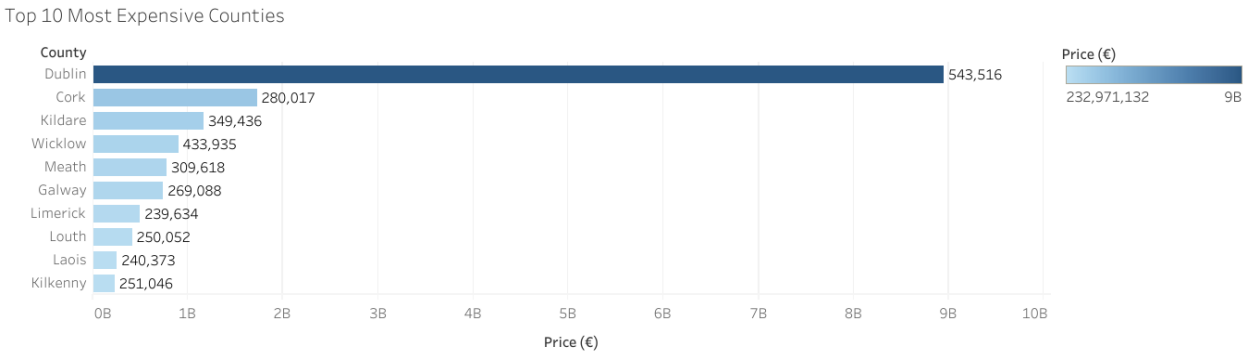


Figure 4 Horizontal bar chart showing the top 10 most expensive counties based on average property values.

2.5 Interactive Elements

To provide granular data without cluttering the visual field, customized interactive tooltips were implemented across all visualizations, enabling users to access precise figures only when needed. This reflects Shneiderman (1996) Visual Information-Seeking Mantra of "overview first, zoom and filter, then details-on-demand," and helps reduce cognitive load by preventing excessive on-screen labels. A global dashboard action was also created, allowing selections on the map or ranking charts to filter the entire dashboard dynamically. This enables users to examine county-specific monthly trends or compare property-type price gaps, supporting deeper synthesis across dimensions. Additionally, a categorical filter for "Description of Property" allows users to toggle between new and second-hand dwellings, facilitating exploratory data analysis within sub-sectors of the market. The inclusion of these interactive features addresses the dataset's complexity while maintaining visual clarity and analytical flexibility for different stakeholders (Shneiderman, 1996).

3 Key Findings and Appraisal

An analysis of the choropleth map (Figure 1) and the horizontal bar chart (Figure 4) clearly show that average property prices in Ireland are strongly concentrated around Dublin and the eastern region. Dublin records the highest average price at €543,516, significantly exceeding all other counties, and appears as the darkest area on the choropleth map. Counties surrounding Dublin — particularly

Wicklow, Kildare, and Meath also display relatively high average prices and darker shades, indicating a clear geographic clustering of higher property values along the east coast. This pattern shows the influence of the capital, where employment opportunities, infrastructure, and population density drive housing demand, with nearby commuter counties benefiting from their proximity.

In contrast, counties in the west and northwest of Ireland appear in lighter shades on the map, indicating lower average property prices. The difference between the highest and lowest averages (approximately €543,516 versus around €141,495) indicates substantial regional disparities in the housing market. Overall, the visualizations suggest that property values decline progressively with distance from Dublin, demonstrating the dominant role of the capital in shaping national price patterns.

The bar chart (Figure 3) comparing the average price of second-hand dwellings and new dwellings shows a substantial price gap between the two property types. New dwellings have a significantly higher average price (approximately €480,000) compared to second-hand dwellings (around €310,000). This difference suggests that buyers pay a considerable amount for newly built properties, likely due to modern construction standards, improved energy efficiency, contemporary design, and lower expected maintenance costs. The clear disparity indicates strong demand for new housing stock and highlights affordability challenges, as new homes may be financially out of reach for many buyers relative to second-hand properties.

The line chart (Figure 2) showing the monthly property sales trend reveals noticeable fluctuations in transaction volumes throughout 2021. Sales increase sharply from January to March, reaching a peak in early spring, before declining in April and May. From mid-year onward, transactions rise again, with another peak occurring around July and remaining relatively high into September. This pattern suggests seasonality in the housing market, where activity tends to strengthen in spring and summer months. Overall, despite short-term dips, the general trend shows that there is sustained and robust market activity across the year.

Taking all these together, the dashboard (Figure 5) reveals that the Irish housing market is characterized by strong geographic concentration, property-type price

disparities, and sustained transactional activity. Property values are highest in and around Dublin, with a clear east–west divide indicating substantial regional inequality in average prices. New dwellings command a significant premium over second-hand properties, suggesting strong demand for modern housing stock and highlighting affordability pressures within the market. At the same time, transaction volumes remain consistently robust throughout the year, with seasonal fluctuations but no prolonged downturn, indicating steady demand overall. Collectively, these patterns point to a dynamic yet uneven property market shaped by location, housing type, and persistent buyer activity.

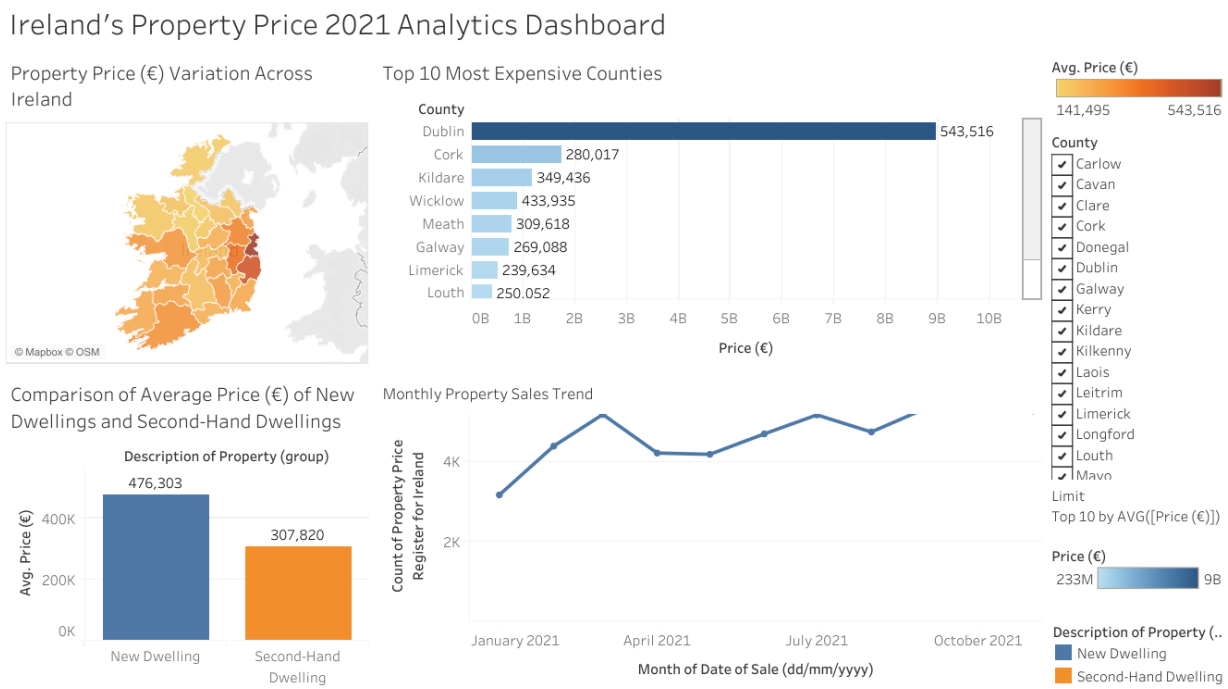


Figure 5 Analytics dashboard for Ireland's Property Price 2021 dataset

4 Data and Process Critique

While the PPR is a robust official record, its utility is restricted by several key metadata limitations that impact the depth of the final conclusions. Although the dataset provides verified transaction prices, dates of sale, property descriptions, and geographic identifiers, it lacks several important contextual variables that would enable a more nuanced interpretation of housing market dynamics. As a result, the insights generated from the analysis, while valid within the dataset's structure, must be interpreted with caution.

One of the most significant limitations is the absence of property size or structural

characteristics. The PPR does not indicate whether a dwelling is a one-bedroom apartment, a three-bedroom semi-detached house, or a five-bedroom detached property. It also does not provide square footage, floor area, number of bathrooms, or land size. Consequently, the calculation of "average price" may be misleading, as it aggregates fundamentally different property types into a single metric. A county with a high average price may simply have a larger proportion of large, detached houses rather than genuinely higher price levels per square metre. Without standardizing prices by size, it is not possible to determine whether differences reflect market demand, housing composition, or structural characteristics. This restricts deeper affordability analysis and limits comparability across counties.

Data cleaning challenges also highlight structural weaknesses in the dataset. During preprocessing, duplicate entries arising from Irish-language translations of property descriptions had to be identified and translated to their corresponding English equivalents. These duplicates, if left unaddressed, would have potentially distorted summary statistics. Additionally, price values required formatting adjustments, including the removal of currency symbols and non-numeric characters before numerical analysis could be conducted. The need for manual cleaning introduces potential for human error and affects reproducibility if not carefully documented.

Another limitation relates to exclusions within the dataset. The PPR does not comprehensively capture certain non-market transactions, including some family transfers, inheritances, or commercial-to-residential property conversions. As a result, the dataset reflects recorded market transactions rather than the full spectrum of property ownership changes. This means that conclusions drawn from the analysis primarily represent open-market activity and may understate housing supply dynamics in specific contexts. Furthermore, the absence of socio-economic indicators such as buyer demographics, mortgage status, or income levels; prevents deeper investigation into affordability pressures or investment behaviour.

Overall, while the PPR dataset provides a strong foundation for examining price patterns and transaction trends, its structural and metadata limitations constrain interpretative depth. The findings generated from the dataset offer valuable high-level insights into geographic disparities, property-type differences, and market activity, but they

cannot fully explain the underlying drivers of price variation.

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